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Advanced Computer Architecture

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Name
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Do ALL (if you know the answers) the theoretical questions but pick 2 out of the 3 problems, DO NOT DO ALL THE 3!
60' to complete the exam.

Questions (20%)	
Problem 1 (40%)	
Problem 2 (40%)	
Problem 3 (40%)	
Total (100%)	

Question 1

Name dependent instructions (Name dependence occurs when 2 instructions use the same register or memory location) can execute simultaneously but their relative order of execution cannot be changed.

(True/False + Effectively support your answer)

Question 2

Theoretically, an unlimited number of issues will be the best option to achieve the highest performance

(True/False + Effectively support your answer)

Question 3

Increasing the number of stages in a pipeline, it is always improving the performance

(True/False + Effectively support your answer)

Question 4

It is not possible to have heterogeneous multicore architectures sharing the same ISA

(True/False + Effectively support your answer)

Question 5

An Embarrassingly Parallel Algorithms is always gaining benefits by an increment in the number of cores provided by the underlying architecture.

Note, in parallel programming, an embarrassingly parallel algorithm is one that requires no communication or dependency between the processes.

(True/False + Effectively support your answer)

Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2	1	2	7	8
I2: ADDD F2 F6 F4	2	9	11	12
I3: MULTD F0 F6 F2	3	13	43	44
I4: SUBD F2 F2 F6	13	14	16	17
I5: ADDD F0 F12 F2	45	46	48	49

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

Problem 2

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 2000, R1 is set to 0)

```
LOOP:    LD      F1  0   R0
          ADDD    F2  F1  F1
          ADDI    R1  R1  1
LOOP2:   MULTD   F2  F2  F1
          SUBI    R1  R1  1
          BNEZ    R1  LOOP2
          SUBI    R0  R0  4
          BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Problem 3

Assume that the following code is executed on a CPU with TOMASULO and with the following units:

- 1 LOAD/STORE unit with Latency equal to 1
- 2 MULT units with latency equal to 10
- 3 ADD/SUBD unit with latency equal to 2

And the following information:

First load 5 clock cycles (cache miss)

Following loads 1 clock cycle (hit)

R1 = 10

Show the computation steps of the following lines of code:

```
Loop:    LD F0 0 R1
         ADDD F6 F4 F0
         MULTD F10 F0 F4
         ADDD F4 F10 F2
         SUBI R1 R1 #5
         BNEZ R1 Loop
```

[illegible]